IMPERIAL COLLEGE LONDON

EE4-25
EE9-CS1-2
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DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

MSc and EEE/EIE PART IV: MEng and ACGI

Corrected copy

DESIGN OF LINEAR MULTIVARIABLE CONTROL SYSTEMS

Monday, 13 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : I.M. Jaimoukha
Second Marker(s) : E.C. Kerrigan

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1. Consider the feedback configuration in Figure 1. Here, $G(s)$ is a nominal plant model and $K(s)$ is a compensator. The stable transfer matrices $\Delta_a(s)$ and $\Delta_m(s)$ represent additive and multiplicative uncertainties on the nominal model.

The design specification are to synthesize a compensator $K(s)$ such that the feedback loop is internally stable when:

(i) $\Delta_a = 0$ and $\|\Delta_m(j\omega)\| \leq |w_m(j\omega)|, \forall\omega$, and,

(ii) $\Delta_m = 0$ and $\|\Delta_a(j\omega)\| \leq |w_a(j\omega)|, \forall\omega$,

where $w_a(s)$ and $w_m(s)$ are appropriate weighting functions.

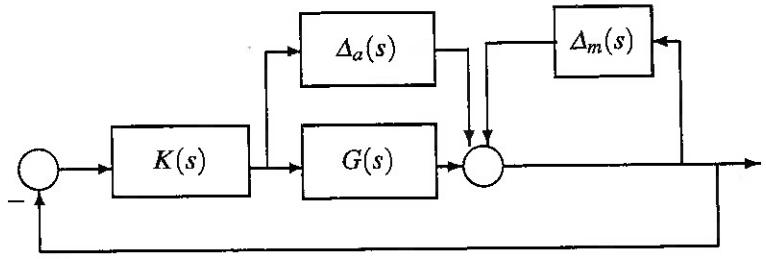


Figure 1

a) State the small gain theorem concerning the internal stability of a feedback loop having a forward transfer matrix Δ and a feedback transfer matrix S. [4]

b) Derive $\mathscr{H}_\infty$–norm bounds, in terms of $G(s)$, $K(s)$, $w_a(s)$ and $w_m(s)$ that are sufficient to achieve the robust design specifications for Figure 1. [4]

c) Define suitable cost signals $z_1(s)$ and $z_2(s)$, external signal $w(s)$, measured signal $y(s)$ and control signal $u(s)$ and draw a block diagram, showing all these signals, as well as suitable weighting functions. [4]

d) Hence derive a generalised regulator formulation of the design problem that captures the sufficient conditions. [4]

e) Assume that $K(s)$ achieves the design specifications in Parts (b)-(d) above. Comment on the tracking properties of the feedback loop in Figure 1. [4]

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2. a) Consider a state–variable model described by the dynamics

$$\dot{x}(t) = Ax(t) + Bu(t), \qquad y(t) = Cx(t) + Du(t),$$

and let $H(s) = D + C(sI - A)^{-1}B$ denote the corresponding transfer matrix. Suppose there exists a $P = P^T \succ 0$ such that

$$\begin{bmatrix} A^TP + PA + C^TC & PB + C^TD \\ B^TP + D^TC & D^TD - \gamma^2 I \end{bmatrix} \prec 0.$$

i) Prove that A is stable. [4]

ii) By defining suitable Lyapunov and cost functions and completing a square, prove that

$$\|H\|_\infty < \gamma.$$

[4]

iii) By using a Schur complement argument, derive a set of conditions that are linear in C and D for $H(s)$ to be stable and $\|H\|_\infty < \gamma$. [4]

b) Consider a state–variable model subject to disturbances $w_1(t)$ and $w_2(t)$ described by the dynamics

$$\dot{x}(t) = Ax(t) + Bw_1(t) + u(t), \qquad e(t) = Cx(t) + w_2(t).$$

An output–injection controller of the form $u(t) = Le(t)$ is to be designed such that, for a given $\gamma > 0$, $\|T_{ew}\|_\infty < \gamma$ where $T_{ew}(s)$ denotes the transfer matrix from $w = \begin{bmatrix} w_1(t) \\ w_2(t) \end{bmatrix}$ to e.

i) Derive a state space realization for $T_{ew}(s)$. [4]

ii) By using the answers to Part (a) above, or otherwise, derive sufficient conditions for the existence of a feasible L. Your conditions should be in the form of the existence of certain solutions to linear matrix inequalities. [4]

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3. Consider the regulator in Figure 3 for which it is assumed that B has full column rank, the pair (A,B) is controllable and $x(0)=x_0$. A stabilizing state-feedback gain matrix F is to be designed such that the cost function

$$J=\left\|\begin{bmatrix} u \\ x \end{bmatrix}\right\|_2^2$$

is minimized.

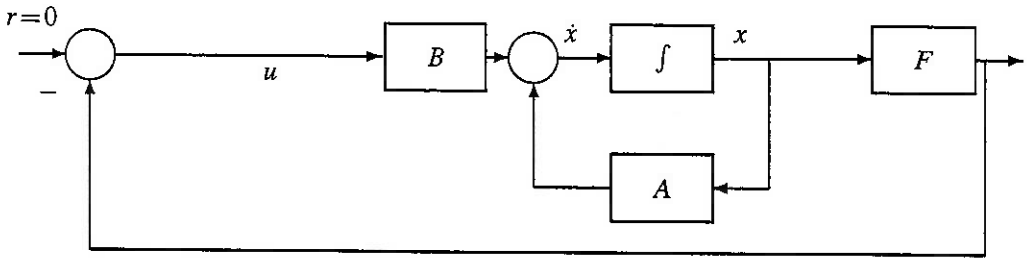


Figure 3

a) By defining suitable cost and Lyapunov functions and completing a square, find an expression for F that achieves the design specifications in terms of the solution of an algebraic Riccati equation. Give the minimum value of the cost function as well as the algebraic Riccati equation. [10]

b) Prove that, for F chosen in part (a), the closed loop system in Figure 3 is stable. State clearly the assumptions required to guarantee stability. [5]

c) Prove that, for F chosen in part (a), the loop gain in Figure 3 is minimum-phase. [5]

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4. Consider the $\mathcal{H}_\infty$ regulator shown in Figure 4 for which it is assumed that the pair (A,B) is controllable and $x(0)=x_0$. Let $H(s)$ denote the transfer matrix from w to $z=[\,z_1^T\ \ z_2^T\,]^T$ and let $\gamma > 0$ be given.

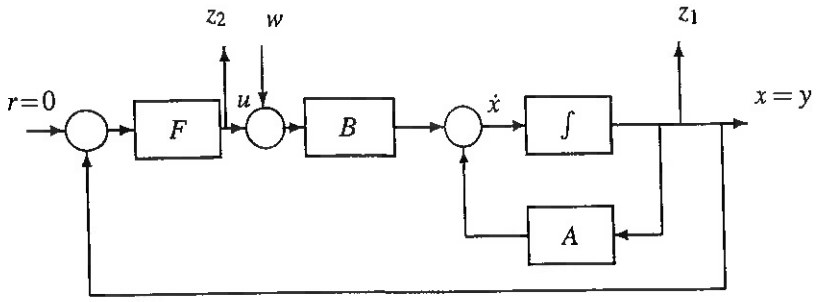


Figure 4

A stabilizing state-feedback gain matrix F is to be designed such that $\|H\|_\infty \leq \gamma$ for a given $\gamma > 0$.

a) Write down the generalized regulator formulation for this design problem. [5]

b) By defining a suitable Lyapunov function and a cost function J involving an auxiliary variable $X = X'$, and carrying out two completions of squares, derive an expression for J that can be used to solve the design problem. [5]

c) Use the expression for J to formulate the design problem and derive sufficient conditions for its solution. Your conditions should be in the form of the existence of a certain solution to an algebraic Riccati equation. It should also include an expression for F and an expression for the worst–case disturbance w^*. [5]

d) Let $\gamma \to \infty$. Write down the cost function and the Riccati equation and compare the solution of the $\mathcal{H}_\infty$ regulator to the solution of the LQR problem in Question 3. [5]

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